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COURSE NAME: FULL STACK WEB DEVELOPMENT

CODE: 23CM4121

TITLE OF THE EXPERIMENT:

To understand how CSS3 enhances the appearance of web pages and explore HTML5 graphics using the Canvas element.

DESCRIPTION(TASK-2):-

This task was carried out to develop a practical understanding of HTML5 graphics using the Canvas element and to explore the visual styling capabilities offered by CSS3, through the creation of a webpage titled "My Styled Webpage." In the first part, the <canvas> element was employed as a drawing surface, controlled entirely through JavaScript, to render a rectangle, a circle, a straight line, and text displaying the student's name, using coordinate-based drawing functions such as fillRect(), arc(), moveTo(), lineTo(), and fillText(). In the second part, the webpage layout was designed to include a header, a navigation bar, three content cards, and a footer, styled comprehensively using CSS3. Various styling properties were applied, including background colours, a linear gradient background, multiple font families and sizes, borders, rounded corners using border-radius, and appropriate margin and padding for spacing. Visual depth was introduced through box-shadow and text-shadow properties, enhancing the overall appearance of the cards and header text respectively. Interactive hover effects were implemented for both the navigation links and the button element, providing immediate visual feedback to the user upon mouse interaction. Overall, the task provided a comprehensive understanding of how HTML5 Canvas facilitates dynamic graphic rendering, and how CSS3 properties collectively contribute to a polished, professional, and user-friendly webpage design.

PROGRAM:-

Task2.html

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>My Styled Webpage</title>

<link rel="stylesheet" href="style.css">

</head>

<body>
```

```

<!-- ===== PART A: CANVAS GRAPHICS ===== -->

<div class="canvas-section">

    <h1>Canvas Graphics</h1>

    <canvas id="myCanvas" width="400" height="250"></canvas>

</div>

<!-- <canvas> is just an empty rectangular box on the page.

    Nothing is drawn yet – it has NO drawing ability of its own.

    Everything visible inside it is drawn using JavaScript. -->


<script>

    const canvas = document.getElementById("myCanvas");
    const ctx = canvas.getContext("2d");

    // getContext("2d") gives us the "drawing tool" (called the context)
    // that has all the draw functions like fillRect, arc, etc.
    // Without this line, none of the drawing commands below would work.


    // ----- RECTANGLE -----
    ctx.fillStyle = "green";
    // fillStyle sets the COLOR used for the next shape drawn
    ctx.fillRect(20, 20, 100, 60);
    // fillRect(x, y, width, height)
    // x,y = top-left corner position of the rectangle
    // width,height = size of the rectangle


    // ----- CIRCLE -----
    ctx.beginPath();
    // beginPath() starts a NEW shape – without this, the circle
    // would connect to the previous shape's path by mistake
    ctx.arc(220, 60, 40, 0, Math.PI * 2);
    // arc(x, y, radius, startAngle, endAngle)
    // 0 to Math.PI*2 means "go all the way around" = a full circle
    ctx.fillStyle = "blue";
    ctx.fill();
    // fill() actually paints the shape with the current fillStyle


    // ----- LINE -----

```

```

ctx.beginPath();
ctx.moveTo(20, 150);
// moveTo(x, y) moves the "pen" to a starting point WITHOUT drawing
ctx.lineTo(300, 150);
// lineTo(x, y) draws a straight line from the last point to this one
ctx.stroke();
// stroke() actually renders the line (moveTo/lineTo alone draw nothing)

// ----- TEXT (Name) -----
ctx.font = "20px Arial";
ctx.fillStyle = "black";
ctx.fillText("Niharika", 20, 200);
// fillText(text, x, y) - x,y is the position of the BOTTOM-LEFT
// of the text, not the top-left (a common beginner mistake)
</script>

```

```

<!-- ===== PART B: CSS STYLING DEMO ===== -->

```

```

<header class="site-header">My Styled Webpage</header>

```

```

<!-- <header> = semantic tag for the top banner section of the page -->

```

```

<nav class="navbar">

```

```

  <!-- <nav> = semantic tag telling the browser "this block is for navigation" -->

```

```

  <a href="#">Home</a>

```

```

  <a href="#">About</a>

```

```

  <a href="#">Contact</a>

```

```

</nav>

```

```

<div class="cards">

```

```

  <!-- A plain <div> is used here since "cards" is just a layout
        container, not a meaningful content type on its own -->

```

```

  <div class="card">

```

```

    <h3>HTML</h3>

```

```

    <p>HTML (HyperText Markup Language) provides the basic structure and content of a webpage using elements like
    headings, paragraphs, images, and links.</p>

```

```

    <button>Learn More</button>

```

```

  </div>

```

```

<div class="card">

    <h3>CSS</h3>

    <p>CSS (Cascading Style Sheets) styles web pages by adding colors, fonts, layouts, spacing, animations, and responsive designs for a better appearance.</p>

    <button>Learn More</button>

</div>


<div class="card">

    <h3>JavaScript</h3>

    <p>JavaScript adds interactivity to web pages by handling events, updating content dynamically, validating forms, and creating animations.</p>

    <button>Learn More</button>

</div>

</div>


<footer class="site-footer">&copy; 2026 My Website</footer>

</body>

</html>

```

style.css

```

html, body{

    height:100%;

    margin:0;

    padding:0;

}


body{

    display:flex;

    flex-direction:column;

    min-height:100vh;

    background:#f5f5f5;

}/* ===== HEADER ===== */

.site-header {

    background: linear-gradient(to right, #345979, #00b8c2);

    /* linear-gradient() blends two colors smoothly into one another,

       instead of the header being one flat, boring color */

    color: white;

    font-family: 'Segoe UI', sans-serif;

```

```

/* font-family sets the TYPEFACE – a different, more modern
   font than the browser's plain default */
font-size: 28px;
text-align: center;
padding: 20px;
/* padding here pushes the text away from the header's own edges,
   so the text isn't jammed right against the border */
text-shadow: 2px 2px 4px rgba(0,0,0,0.4);
/* text-shadow draws a soft shadow BEHIND the letters,
   making white text stand out more against the gradient */
}

```

```

.canvas-section{
    width:90%;
    text-align:center;
    margin:30px auto;
}

```

```

canvas{
    width:100%;
    max-width:500px;
    height:350px;
    border:2px solid black;
}

```

```

/* ===== NAVIGATION BAR ===== */

```

```

.navbar {
    background-color: #f0f0f0;
    padding: 10px;
    text-align: center;
}

```

```

.navbar a {
    color: #333;
    margin: 0 15px;
    /* margin here creates gaps BETWEEN each link, since margin is

```

```

    space OUTSIDE an element, pushing neighbours away */
text-decoration: none;

font-size: 18px;

font-family: Georgia, serif;

/* a different font family used here vs. the header,
    to satisfy "different font families and sizes" */
}

.navbar a:hover {
    color: #ff6600;

    /* :hover = a pseudo-class, styles applied ONLY while the
        mouse pointer is over the link – this is the "hover effect" */
}

/* ===== CONTENT CARDS ===== */
.cards {
    display: flex;

    justify-content: center;

    gap: 20px;

    margin: 30px;
}

.card {
    background-color: #f4f4f4;

    border: 1px solid #ccc;

    /* border is part of the box's own outline – takes up layout space */

    border-radius: 12px;

    /* border-radius rounds off the sharp corners of the box */

    padding: 30px;

    /* padding here creates space INSIDE the card, between its
        border and the heading/paragraph/button inside it */

    width: 300px;

    height: 200px;

    box-shadow: 3px 3px 10px rgba(0,0,0,0.2);

    /* box-shadow adds a soft shadow around the whole card,
        giving it a slight "lifted off the page" depth effect */
}

```

```

    text-align: center;
}

/* ===== BUTTON ===== */
button {
    padding: 10px 20px;
    border: none;
    border-radius: 6px;
    background-color: #4facfe;
    color: white;
    cursor: pointer;
    /* cursor:pointer changes the mouse icon to a hand,
       signalling "this is clickable" */
}

button:hover {
    background-color: #0077cc;
    /* hover effect: background color changes when the mouse is over it */
}

/* ===== FOOTER ===== */
.site-footer {
    width: 100%;
    background-color: #222;
    color: white;
    text-align: center;
    padding: 15px;
    margin-top: auto;
    /* margin-top pushes the footer away from whatever content
       is above it, keeping the layout from feeling cramped */
}

```

OUTPUT:-

